

20251202

Rem $f: U \subseteq V \rightarrow W$

$\uparrow$
 open set

$\uparrow$
 vector spaces

$$A \mapsto f(A)$$

we mentioned that for fixed $A \in U \subseteq V$

$$Df(A): V \rightarrow W,$$

$$H \mapsto Df(A) \cdot H$$

is linear.

The derivative function $Df = f'$ is

$$Df: U \times V \rightarrow W \quad (\text{Fréchet Derivative})$$

$$(A, H) \mapsto Df(A) \cdot H$$

or $Df: U \rightarrow (V \rightarrow W) \quad (\text{currying})$

$$A \mapsto Df(A)$$

$$(H \mapsto Df(A) \cdot H)$$

$$D(f \circ g)(x) = Df(g(x)) = Dg(x)$$

$$\frac{d}{dt} \det(A(t))$$

$$\left(\begin{aligned} &= \frac{d}{dt} (\det \circ A)(t) \times \text{by 1} \\ &= \frac{d}{dt} (\det \circ A)(t) \cdot 1 \\ &= D(\det \circ A)(t) \cdot 1 \end{aligned} \right)$$

$$\det \circ A: \mathbb{R} \rightarrow \mathbb{R}$$

$$\begin{array}{ccccc} & & \text{det} & & \\ & \text{A}(\cdot) & \xrightarrow{\quad} & M_n(\mathbb{R}) & \xrightarrow{\quad} & \mathbb{R} \\ \mathbb{R} & \xrightarrow{\quad} & & & & \\ t & \mapsto & A(t) & \mapsto & \det(A(t)) & \end{array}$$

$$= (D \det(A(t)) \circ DA(t)) \cdot 1 \quad (f \circ g)(v) = f(g(v))$$

↑ composition of linear maps

$$= D \det(A(t)) \cdot (DA(t) \cdot 1)$$

$$= D \det(A(t)) \cdot \frac{dA(t)}{dt}$$

$$\begin{aligned} (A \circ B)(v) \\ &= A(Bv) \\ &= A \cdot B \cdot v \end{aligned}$$

$$= D \det(a_1(t), a_2(t), \dots, a_n(t)) \cdot (\dot{a}_1(t), \dots, \dot{a}_n(t))$$

↑
first col.
of $A(t)$

↑
first column
of $\dot{A}(t) = \frac{dA(t)}{dt}$

$$= \det(\dot{a}_1(t), a_2(t), \dots, a_n(t))$$

$$+ \det(a_1(t), \dot{a}_2(t), a_3(t), \dots, a_n(t))$$

$$+ \dots + \det(a_1(t), \dots, a_{n-1}(t), \dot{a}_n(t))$$

$$= \sum_{k=1}^n \det(a_1(t), \dots, a_{k-1}(t), \dot{a}_k(t), a_{k+1}(t), \dots, a_n(t))$$

Lemma Let $A \in M_n(\mathbb{R})$. then

$$D \det(I_n) \cdot A = \text{tr}(A)$$

Indeed, $D \det(I_n) \cdot A$

$$= D \det \left(\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \right) \cdot \left(\begin{bmatrix} 1 \\ a_1 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \right)$$

$$= \det \underbrace{\begin{pmatrix} a_{11} & 0 & 0 & \dots & 0 \\ a_{21} & 1 & 0 & \dots & 0 \\ \vdots & 0 & 1 & \dots & 0 \\ a_{n1} & 0 & 0 & \dots & 1 \end{pmatrix}}_{a_{11}} + \det \underbrace{\begin{pmatrix} 1 & a_{12} & 0 & \dots & 0 \\ 0 & a_{22} & 0 & \dots & 0 \\ 0 & a_{32} & 1 & \dots & 0 \\ \vdots & \vdots & 0 & \ddots & \vdots \\ 0 & a_{n2} & \vdots & \vdots & 1 \end{pmatrix}}_{a_{22}}$$

$$+ \dots + \det \underbrace{\begin{pmatrix} 1 & 0 & \dots & 0 & a_{1n} \\ 0 & 1 & & & a_{2n} \\ \vdots & 0 & \ddots & & 0 \\ 0 & 0 & \dots & 1 & a_{nn} \end{pmatrix}}_{a_{nn}}$$

$$= a_{11} + a_{22} + \dots + a_{nn} = \text{tr}(A)$$

□

Prop Let $A \in GL_n(\mathbb{R})$, then for $H \in M_n(\mathbb{R})$

$$D \det(A) \cdot H = \det(A) \underbrace{\text{tr}(A^{-1}H)}_{\text{tr}(HA^{-1})}$$

Pf 1 By def'n

$$D \det(A) \cdot H = \left. \frac{d}{dt} \right|_{t=0} \det(A + tH)$$

$$= \left. \frac{d}{dt} \right|_{t=0} \det(A) \det(I_n + tA^{-1}H)$$

$$(\det(AB) = \det(A) \det(B))$$

$$= \det(A) \underbrace{\left. \frac{d}{dt} \right|_{t=0} \det(I_n + tA^{-1}H)}_{\downarrow \text{def'n}}$$

$$= \det(A) [D \det(I_n) \cdot (A^{-1}H)]$$

$$= \det(A) \text{tr}(A^{-1}H)$$

□

Pf 2 $\det : GL_n(\mathbb{R}) \rightarrow GL_1(\mathbb{R}) \cong \mathbb{R} \setminus \{0\}$

$$\begin{array}{ccc}
 (A, B) & \xrightarrow{x \text{ in } GL_n(\mathbb{R})} & AB \\
 (\det, \det) \downarrow & & \downarrow \\
 (\det A, \det B) & \xrightarrow{x \text{ in } \mathbb{R} \setminus \{0\}} & \det(AB) = \det(A) \det(B)
 \end{array}$$

$$\begin{array}{ccc}
 \det(AB) & = & \det(A) \det(B) \\
 \uparrow & & \uparrow \\
 x \text{ in } & & x \text{ in } \\
 GL_n(\mathbb{R}) & & GL_1(\mathbb{R}) \cong \mathbb{R} \setminus \{0\}
 \end{array}$$

Treat this as some function of A applied to B

$$\Leftrightarrow \det(L_A(B)) = l_{\det(A)}(\det(B))$$

where

$$\text{linear} \left\{ \begin{array}{ll} L_A : M_n(\mathbb{R}) \rightarrow M_n(\mathbb{R}) & \Rightarrow DL_A(x) = L_A \quad \forall x \in M_n(\mathbb{R}) \\ B \mapsto AB \\ l_\alpha : \mathbb{R} \rightarrow \mathbb{R} & \Rightarrow Dl_\alpha(x) = l_\alpha \quad \forall x \in \mathbb{R} \setminus \{0\} \\ x \mapsto \alpha x \end{array} \right.$$

$$Dl_\alpha(x) \cdot h = \alpha h$$

$$\Leftrightarrow \det \circ L_A = l_{\det(A)} \circ \det$$

Then

$$\begin{aligned} D \det(A) \cdot H &= D \det(A) \cdot A (A^{-1}H) \\ &= D \det(A) \cdot L_A(A^{-1}H) \\ &= D \det(\underbrace{A}_{L_A(I_n)}) \cdot D \underbrace{L_A(I_n)}_{L_A(I_n)} \cdot A^{-1}H \\ &= D(\det \circ L_A)(I_n) \cdot A^{-1}H \\ &= D(\ell_{\det A} \circ \det)(I_n) \cdot A^{-1}H \\ &= D \ell_{\det A}(\det(I_n)) \cdot (D \det(I_n) \cdot A^{-1}H) \\ &= \ell_{\det A}(\text{tr}(A^{-1}H)) \\ &= (\det A) \text{tr}(A^{-1}H) \end{aligned}$$

□

$$L_A : M_n(\mathbb{R}) \longrightarrow M_n(\mathbb{R})$$

$$X \longmapsto AX \quad \Leftrightarrow \quad L_A(X) = AX$$

$$D L_A(X) \cdot H = \left. \frac{d}{dt} \right|_{t=0} L_A(X + tH)$$

$$= \left. \frac{d}{dt} \right|_{t=0} A(X + tH) = \left. \frac{d}{dt} \right|_{t=0} (AX + t \underline{AH}) = AH$$

$$\text{So } D L_A(X) : M_n(\mathbb{R}) \longrightarrow M_n(\mathbb{R}), \quad H \longmapsto AH$$

$$D L_A(X) \cdot H = L_A(H) \quad \forall X, H \in M_n(\mathbb{R})$$

$$\text{or } D L_A(X) = L_A \quad \forall X \in M_n(\mathbb{R})$$

Thm Let $A \in M_n(\mathbb{R})$, then

$$\begin{array}{ccc} \det(e^{tA}) & = & e^{t(\operatorname{tr} A)} \quad \forall t \in \mathbb{R} \\ \uparrow & & \uparrow \\ \exp: M_n(\mathbb{R}) & & \exp: \mathbb{R} \rightarrow \mathbb{R} \\ & \searrow & \\ & M_n(\mathbb{R}) & \end{array}$$

Pf We verify that both sides satisfy the same IVP

at $t=0$

$$\text{LHS} = \det(e^{0A}) = \det\left(I_n + 0A + \frac{(0A)^2}{2!} + \dots\right) = \det(I_n) = 1$$

$$\text{RHS} = e^{0(\operatorname{tr} A)} = e^0 = 1$$

$$\frac{d}{dt} \text{LHS} = \frac{d}{dt} \det(e^{tA}) = \frac{d}{dt} (\det \circ \exp)(tA)$$

$$= D \det(e^{tA}) \cdot \frac{d}{dt} e^{tA}$$

$$= D \det(e^{tA}) \cdot e^{tA} A$$

$$= \det(e^{tA}) \operatorname{tr}(\cancel{e^{-tA}} \cancel{e^{tA}} A)$$

$$= \operatorname{tr}(A) \underbrace{\det(e^{tA})}_{\text{LHS}}$$

$$\frac{d}{dt} \text{RHS} = \frac{d}{dt} e^{t(\operatorname{tr} A)} = \operatorname{tr}(A) e^{t(\operatorname{tr} A)} = \operatorname{tr}(A) \text{RHS}$$

(both satisfy IVP $\dot{X} = \operatorname{tr}(A)X$, $X(0) = 1$)

By uniqueness, $\text{LHS} = \text{RHS}$. $\square$

$$\underline{\text{Ex}} \quad W(t) = \det \bar{\Psi}(t), \quad \&$$

$$\frac{d}{dt} \bar{\Psi}(t) = A(t) \bar{\Psi}(t)$$

$$\text{where } \bar{\Psi}(t) \in M_n(\mathbb{R}) \quad \forall t \in I \subseteq \mathbb{R}$$

Then

$$\frac{d}{dt} W(t) = \frac{d}{dt} \det \bar{\Psi}(t) \quad \left(= \frac{d}{dt} (\det \circ \bar{\Psi})(t) \right)$$

$$= D \det(\bar{\Psi}(t)) \cdot \frac{d}{dt} \bar{\Psi}(t)$$

$$= D \det(\bar{\Psi}(t)) \cdot A(t) \bar{\Psi}(t)$$

$$= \det(\bar{\Psi}(t)) \operatorname{tr} (A(t) \bar{\Psi}(t) \bar{\Psi}(t)^{-1})$$

$$= (\operatorname{tr} A(t)) W(t)$$

This is a scalar ODE, w/ solution

$$W(t) = W(t_0) \exp \left(\int_{t_0}^t \operatorname{tr}(A(s)) ds \right)$$

We recover the previous case if $A(t) \equiv A$ const.